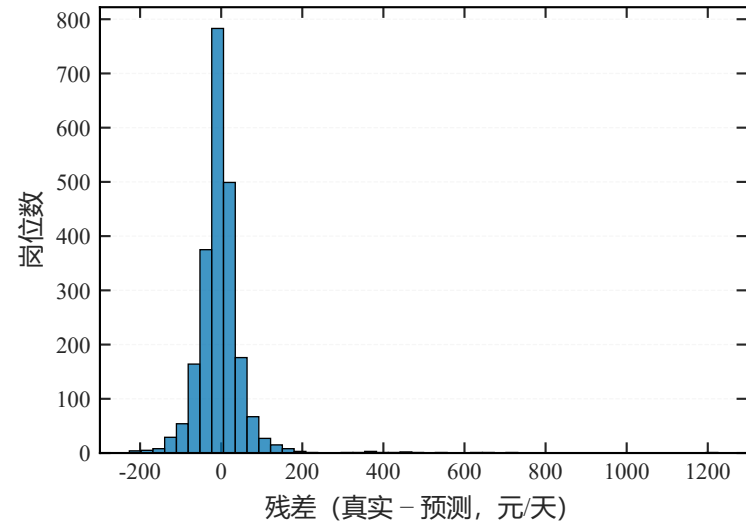
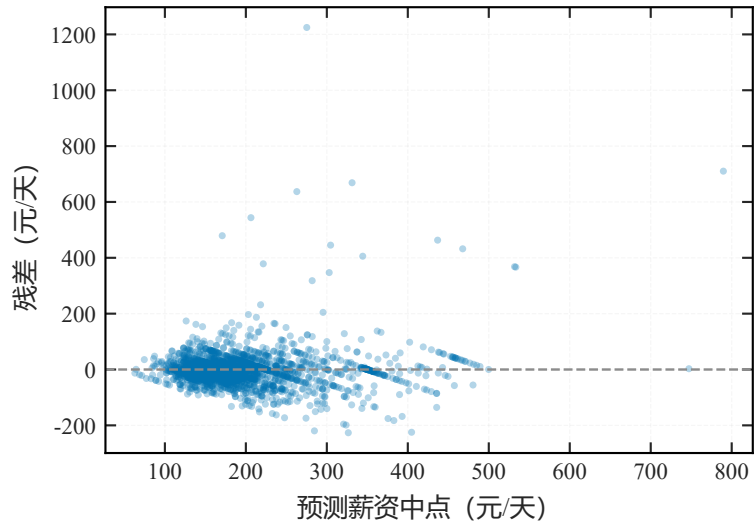




(a) 残差分布直方图



(b) 残差 vs 预测值

图13 正式主模型 (LightGBM) test 残差诊断